

Paper: 1MA1/3H				
Question	Answer	Mark	Mark scheme	Additional guidance
20	$x = 3\frac{2}{5}, y = -\frac{4}{5}$	M1	for substitution of a rearrangement eg $x = \frac{7-4y}{3}$ or $y = \frac{7-3x}{4}$ into $x^2 - 4y^2 = 9$ or expansion of $\left(\frac{7-4y}{3}\right)^2 = \frac{49-56y+16y^2}{9}$ or $\left(\frac{7-3x}{4}\right)^2 = \frac{49-42x+9x^2}{16}$	Expansion may not be in simplest form but must be correct
		M1	for correct expansion and substitution eg $\frac{49-56y+16y^2}{9} - 4y^2 = 9$ or $x^2 - 4\left(\frac{49-42x+9x^2}{16}\right) = 9$	
	$x = 5, y = -2$	A1	for forming quadratic ready for solving eg $-20y^2 - 56y - 32 (= 0)$ or $5y^2 + 14y + 8 (= 0)$ oe or $5x^2 - 42x + 85 (= 0)$ oe	Note we do not need to see “= 0”; just the LHS is sufficient.
		M1	ft a 3 term quadratic , factorising eg $(5y + 4)(y + 2) (= 0)$ or $(5x - 17)(x - 5) (= 0)$ or correct use of formula eg $(y =) \frac{-14 \pm \sqrt{14^2 - 4 \times 5 \times 8}}{2 \times 5}$ or $(x =) \frac{- -42 \pm \sqrt{42^2 - 4 \times 5 \times 85}}{2 \times 5}$ or completing the square, eg $(y + \frac{7}{5})^2 - \frac{9}{25} (= 0)$ or $(x - \frac{21}{5})^2 - \frac{16}{25} (= 0)$	Can be implied by both x values correct or both y values correct.
		A1	correctly pairs x and y values: $x = 3\frac{2}{5}, y = -\frac{4}{5}$ oe , $x = 5, y = -2$	Answers must be correctly paired. Accept coordinate pairs